

Multispectral imaging for predicting the water status in mushroom during hot-air dehydration

Shoaib Younas, Changhong Liu, Hao Qu, Yu Mao , Wei Liu, Liyang Wei, Ling Yan, and Lei Zheng

Abstract: In-depth understanding of the shifting of water status during dehydration is crucial for obtaining better quality of dried food. In this work, we report a nondestructive method to measure the water status in hot-air dried mushroom via multispectral imaging (MSI) technology combined with chemometric methods. The low-field nuclear magnetic resonance (LF-NMR) measurements were performed as reference. During drying process, the moisture content changed dramatically with notable migration and conversion of different water phases. Partial least squares (PLS), back propagation neural network (BPNN), and least squares-support vector machine (LS-SVM) models were applied to develop quantitative models. Among all, BPNN model showed considerably better performance of prediction with coefficient of determination $R^2c = 0.9829$, $R^2p = 0.9639$. The results demonstrated that MSI technology combined with chemometric methods is an impressive approach for determination of the water status in hot-air dried mushrooms, which would facilitate infield of food processing by providing applicable and appropriate platform.

Keywords: hot-air drying, multispectral imaging, mushroom, nondestructive technique, water status

Practical Application: Experimental investigation of different water status during food processing. Assessment of the potential of multispectral imaging to predict water status. Usage of novel measurement method for food processors.

1. INTRODUCTION

Mushroom is well known as a health-stimulating food considering the enrichment of flavors, phenolic compounds, digestible proteins, and essential amino acids, which has made it potentially to be used as a medicine (Chung et al., 2018; Villares, Mateo-Vivaracho, & Guillamón, 2012). United Nations Food and Agriculture Organization stated that about 11 mega tons of mushrooms and truffles were produced in 2016. Asia was a highly saturated region in mushroom production, while China contributed 72% of world's total production (FAOSTAT, 2016). Shiitake mushroom is one of the most cultivated fungus because of its appreciable flavor and nourishment. It has achieved more attention and higher consumption rate due to therapeutic effects on various diseases and disorders such as bacterial infections (Hearst et al., 2009; Maria Jose et al., 2013), inflammation, and cardiovascular problems (Pappetti et al., 2018).

As plant-based foods, fresh mushrooms are porous and contain 80% to 90% of water. The high moisture contents of postharvested mushrooms could promote microbial activity, which would result in rapid deterioration within a day (Antmann, Ares, Lema, & Lareo, 2008; Guo, Xia, Tan, Chen, & Ming, 2014; Walde, Velu, Jyothirmayi, & Math, 2006). The shorter life-span is problematic for marketing, transportation, and storage. Mushrooms are subjected to different water-removal techniques to control microbial growth, prolong life-span, and stop the enzymatic activity of

mushrooms up to an acceptable level (Jaworska, Pogoń, Bernaś, & Skrzypczak, 2014). Water could be categorized into three different types on the mobility basis, namely, free water, immobilized water, and bound water (Matthew, 2011). The investigation of water status provides a route map, which shows whether drying conditions are favorable or not. Various researches have been carried out to measure water status in plant-based materials. Fei et al. (2018) studied the osmotic and ultrasound-assisted osmotic drying conditions using low-field nuclear magnetic resonance (LF-NMR), showing that osmotic drying would cause more vacuole water loss, which led to shrinkage and less efficiency in quality retention compared to ultrasound-assisted drying.

LF-NMR is used for the detection of water status of various food products (Cheng et al., 2018), and the spin-spin relaxation time (T2) signal provides useful information of water proportion and the interaction with macromolecules within plant tissue (Khan, Wellard, Nagy, Joardder, & Karim, 2016). As an example, LF-NMR was used to investigate water status in chicken breast during drying process (Li et al., 2014). Khan et al. (2016) carried out research on fruits and vegetables for the investigation of free, loosely bound, and strongly bound water through ^1H -NMR. The efficient measurement yielded significant figures of different water forms, for example, the proportions of FW (free water), LBW (loosely bound water), and SBW (strongly bound water) are about 80% to 92%, 6% to 16%, and 1% to 5%, respectively. The water fraction and distribution of shiitake mushroom during hot-air drying processing were also determined using LF-NMR (Lu, Guo, Li, & Ming, 2017). Even though the LF-NMR-based approach is believed to be rather effective for precise determination of water status, it is also expensive, complicated, and time consuming. In order to achieve online determination, the development of a rapid, noninvasive, and cost-effective detection method is urgently required.

The multispectral imaging (MSI)/hyperspectral imaging technology has gained widespread acceptance for agro-food, remote

JFDS-2019-0698 Submitted 5/9/2019, Accepted 1/24/2020. Authors Younas, C. Liu, Qu, Mao, Wei, Yan, and Zheng are with School of Food and Biological Engineering, Hefei University of Technology, Hefei 230009, China. Author W. Liu is with Intelligent Control and Compute Vision Lab, Hefei University, Hefei 230601, China. Author Zheng is also with Research Laboratory of Agricultural Environment and Food Safety, Anhui Modern Agricultural Industry Technology System, Hefei 230009, China. Direct inquiries to authors Mao and Zheng (E-mail: maoyu@hfut.edu.cn; lzhang@hfut.edu.cn; lei.zheng@aliyun.com).

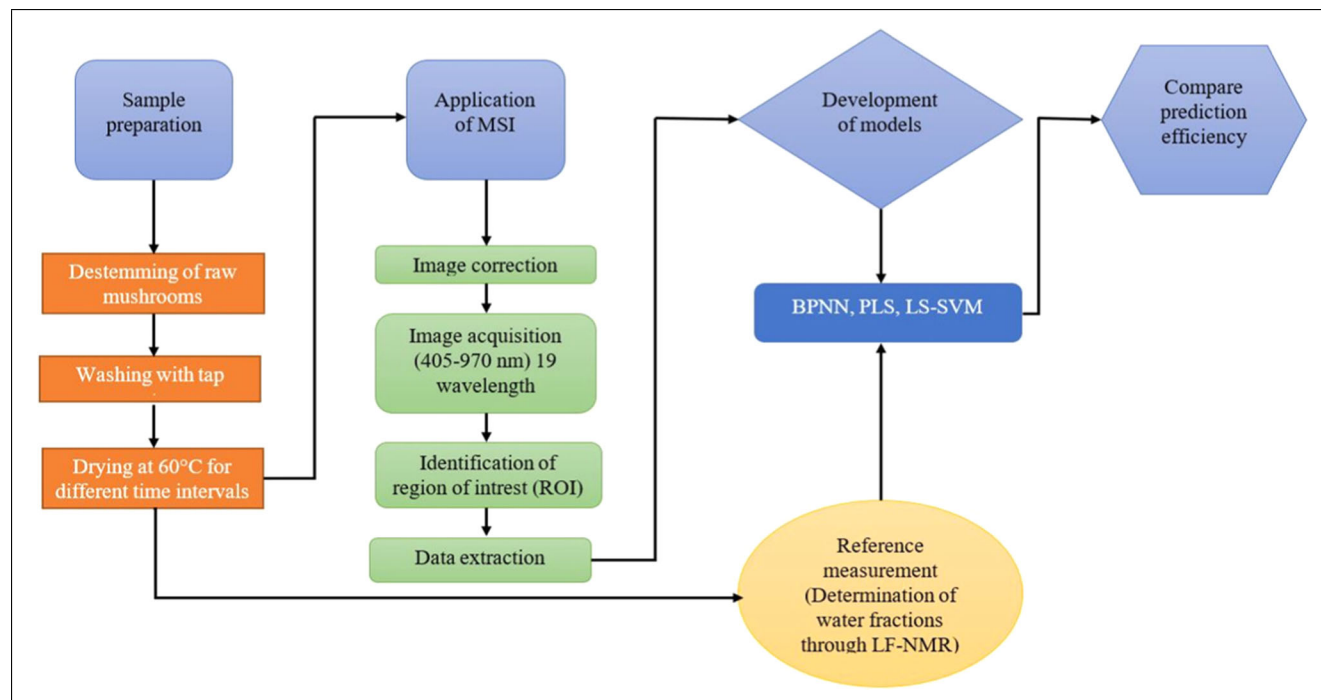


Figure 1—Flow diagram of experimental process.

sensing, and biomedical applications. Recently, it has been used in different studies in quality evaluation of different agro-food products, for example, for predicting heme and nonheme iron contents in pork sausage (Ma et al., 2016), transgenic assessment of crop seeds (Liu et al., 2014a, 2016a), classification of storage period in tea (Xiong et al., 2015), quality evaluation of rocket (Løkke, Seefeldt, Skov, & Edelenbos, 2013), quality monitoring of in-shell infestation in almonds and sunflower seeds (Ma et al., 2015; Yu et al., 2019), characterizing sensory properties and physicochemical parameters in strawberry and tomato fruits (Li et al., 2014; Liu, Liu, Chen, Yang, & Zheng, 2015; Liu et al., 2014b), pork microbial safety identification (Dissing et al., 2013; Ma et al., 2014), and adulteration prediction in meat, tomato paste, and infant formula powder (Liu et al., 2017; Liu, Liu, Yang, Chen, & Zheng, 2017; Ropodi, Pavlidis, Mohareb, Panagou, & Nychas, 2015a). Furthermore, spectral imaging holds high potential for quality assessment relevant to moisture contents, for example, for inspection of frozen minced beef followed by thawing process (Ropodi, Panagou, & Nychas, 2018), identification of moisture contents in carrots (Liu et al., 2016b), and dehydrated prawns (Wu et al., 2012). Although MSI technology is a promising solution, it still remains a challenge to monitor water status in mushroom during drying process. Hereby stating the key objective of this research work was the assessment of MSI application for prediction of water status in mushroom at different stages of hot-air drying and explore the prediction performance of partial least squares (PLS), back propagation neural network (BPNN), and least squares-support vector machine (LS-SVM) models.

2. MATERIALS AND METHODS

The framework of the experimental design (Figure 1) describes four major steps including substeps of present research such as sample preparation, application of MSI, model development, and results of water status.

2.1 Sample preparation

We purchased the raw shiitake mushrooms (*Lentinus edodes*) with uniform size and shape from supermarket Carrefour (Hefei, Anhui Province, China), and the mushrooms were destemmed prior to storage at 4 °C. Before proceeding hot-air drying, tap water was used to remove contaminations and blotted to dry surface water. Mushrooms used for drying were sliced in cube (2 × 2 × 2 cm) dimensions. Initial moisture content (w.b.) of the mushrooms was 91.03% ± 1.93% measured by placing samples in a drying oven (105 °C) until the achieving a constant weight according to the Association of Official Agricultural Chemists (AOAC) method (AOAC, 2000). The drying process was practiced in a lab-scaled hot-air drier (SANFA, Model DHG-9101-2SA Shanghai, China) with a smooth temperature of 60 °C. Multispectral images were captured for mushrooms before exposed to hot-air drying. The water fractions investigation was approached within time intervals of 0, 120, 180, 240, 360, and 420 min. Multispectral images were collected for each time interval before measurement of moisture loss according to following equation:

$$\text{Water contents (\%)} = (M_0 - M_1)/M_0 \times 100 \quad (1)$$

In the above equation, M_0 indicated initial mass and M_1 indicated mass of dried samples at a specific time.

2.2 Image acquisition and analysis

Multispectral images were collected at 19 different nonuniform wavelengths between UV (405 nm) and NIR (970 nm) through the implementation of VideometerLab (software) established by a Denmark institute named Technical University of Denmark (Carstensen & Hansen, 2003). The wavelength details are 405, 435, 450, 470, 505, 525, 570, 590, 630, 645, 660, 700, 780, 850, 870, 890, 910, 940, and 970 nm. A standard monochrome charge coupled device chip was engaged to commemorate the surface reflectance. To avoid scattering and confirmation of uniform

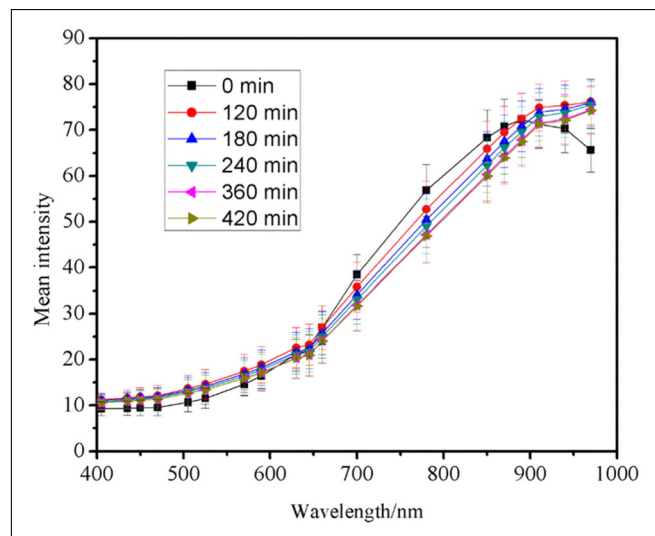


Figure 2—Average reflected spectrum of mushroom samples generated by 300 samples at different dehydration periods.

reflection, mushroom samples were placed on a white coated slide inside an Ulbricht sphere contained with a top-mounted camera. Narrow-banded spectral light emitting diodes (LEDs) are configured in such a way that they generate uniformity inside the entire sphere. The LEDs are switched on at the time of image acquisition and when a spectral reflection of a particular wavelength is detected through the top-mounted camera. The radiometrical and geometrical calibration of the system was accomplished by means of a light setup (auto light) taking into consideration the recognition of object type (Dissing et al., 2013). A preprocessing threshold step followed by image segmentation using VideometerLab software version 2.12.23 enabled the elimination of irrelevant objects from the image, except mushroom tissues (Daugaard, Adler-Nissen, & Carstensen, 2010). To obtain the region of interest (ROI), canonical discriminant analysis (CDA) was applied that removed irrelevant background (Cruz-Castillo et al., 1994). The mean reflectance spectrum was calculated for ROI at each wavelength by mean to take into account the average intensity of pixels.

2.3 Measurement of water mobility

LF-NMR was used to determine the water mobility with the same time intervals during the drying process for the assessment of MSI potential. T_2 relaxation experiments were performed on LF-NMR (Niumag Benchtop Pulsed NMR Analyzer PQ001) installed in Shanghai (Niumag Electric Corporation, Shanghai, China). The resonance frequency for protons of the operated NMR analyzer was 23 MHz at 32 °C with 90° and 180° pulses of 18 μ s and 38 μ s, respectively. Carr–Purcell–Meiboom–Gill (CPMG) sequence was used for determination of spin–spin relaxation time (T_2). All the samples were analyzed by placing them one by one in an NMR glass tube of 25 mm diameter and injected in the NMR probe, with three replications used for each mentioned drying time. The key parameters of T_2 relaxation time, TW (time waiting), TE (time echo), NECH (number of echoes), NS (number of scan), were TW = 2,500 ms, TE = 0.2 ms, NECH = 1,000, NS = 64, respectively. The water proportion of each category was measured at per gram and T_2 (water distribution) plotted by logarithmic coordinates of the original data.

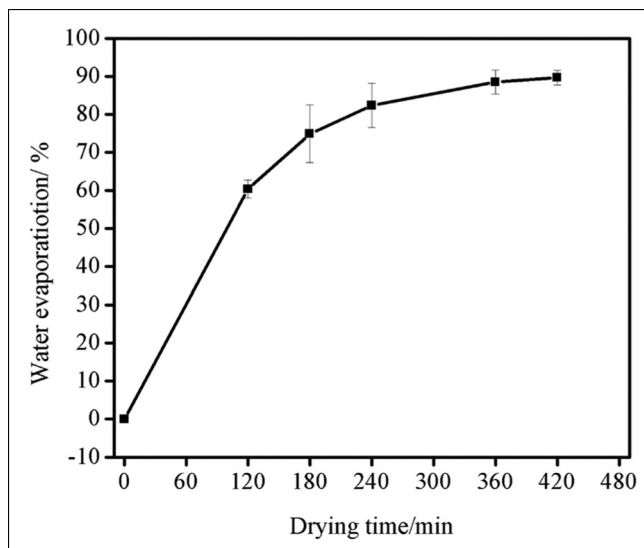


Figure 3—Evaporation of water from shiitake mushroom with different drying periods.

2.4 Establishment of calibration models

The measured water contents and spectral data of different dehydration time periods were analyzed through the establishment of three different quantitative models, namely, PLS, SVM, and BPNN.

A multivariate calibration statistical data analysis approach was achieved by successful implementation of a well-defined bilinear regression PLS model. This model was carried out in fruit and vegetable sector to obtain desirable outcomes. The linear combinations of large variables of spectral data (X) assist to verify the analyte's concentration (Y), precisely compressed in PLS into a much smaller form of latent variables (LVs), and largely explain the covariance of X and Y . Determination of the proper number of LVs tactic was employed in the algorithm to resolve overfitting through minimizing the value of the prediction residual error sum of square (PRESS). In PLS, the X variable entitled as the reflected spectral data, although the Y variable motioned as actual moisture values determined by the oven-drying method.

Cortes and Vapnik (1995) proposed the support vector machine (SVM), which is an effective algorithm designed to accomplish the classification and regression expressions. The efficient capability of SVM is to deal with large input spaces, noisy patterns, and multimodal class distributions, and their limitations on only a subset of training data in order to fit a (nonlinear) function. LS-SVM based on standard SVM is an optimized version focusing on linear equations rather than solving quadratic programming problem. Besides generalization performance, SVM holds the advantage to simplify complex structure and reduce optimization time. Mathematical expression of LS-SVM is shown in following equation:

$$y(x) = \sum_{i=1}^n a_i K(x, x_i) + b \quad (2)$$

In the above equation, $K(x, x_i)$, x_i , and a_i indicate the kernel function, input factor, and Lagrange multipliers, respectively, whereas b is the basic term.

LS-SVM required only two parameters (γ , σ^2), in which γ was used to regulate the process and concerned with reducing training error and model complications while σ^2 designated to

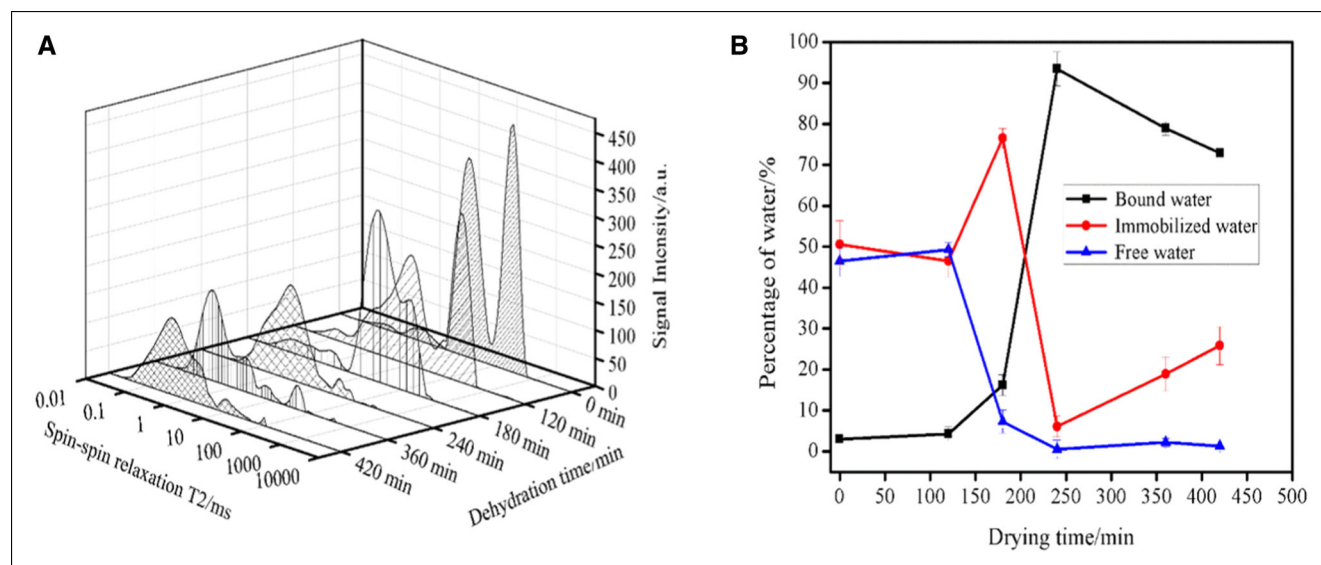


Figure 4—LF-NMR measurement of water fractions: (A) 3D graphical explanation of hot-air effect on different T_2 , that is, free water/ T_{23} (100–1000 ms), immobilized water/ T_{22} (10–100 ms), and bound water/ T_{21} (0.01–10 ms), (B) amount of water fraction of under-process.

Table 1—Descriptive statistics of moisture contents (%) of calibration and prediction sets.

Properties	No. of samples	Minimum	Maximum	Mean	SD
Calibration	198	0.53	93.86	26.66	31.32
Prediction	102	0.28	92.91	22.44	31.39

Table 2—Potential of different calibration models to predict water status of shiitake mushroom.

Model	R^2_c	RMSEC	R^2_p	RMSEP
LS-SVM	0.9596	30.8423	0.7179	31.1673
PLS	0.9546	8.8511	0.9304	15.3749
BPNN	0.9829	0.2289	0.9639	0.2291

spectral width and execution of nonlinear plotting attributes of input region to particular properties of a huge dimensional region. Leave-one-out cross-validation was directed to accomplish the model parameters optimization, which resulted in high accuracy rate. All these chemometrics were conducted in MATLAB (2017; The Math Works Inc., Natick, MA, USA).

2.5 Model performance evaluation

The processed 300 samples of shiitake mushrooms were separated in two sets (calibration and predictions sets) of 2:1 ratio. The evaluation of different established models counted coefficient determination for calibration (R^2_c), coefficient determination for prediction (R^2_p), root-mean square error of calibration (RMSEC), and root-mean square error of prediction (RMSEP). The significant assessment of relationship among models and evaluation were conducted on the basis of ratio between RMSEP and RMSEC. Normally, good model must acquire higher values for R^2_c , R^2_p , RPD, and low values of RMSEC and RMSEP, as well as small difference between RMSEC and RMSEP (C. Liu et al., 2016).

2.6 Statistical analysis

The measurements of all chemometric analyses and plotting were analyzed using MATLAB 2017b (The Math Works Inc.), VideometerLab 2.12.23 (Videometer A/S, Høsholm, Denmark),

and origin 8.0. IBM SPSS Statistics V.23 were used for descriptive analysis.

3. RESULTS AND DISCUSSION

3.1 Reflected spectral analysis

The average reflected spectrum intensity of mushrooms through the MSI system at different drying periods is plotted in Figure 2. There was a continuous rise in reflectance intensity corresponding with the wavelength but relatively lower in each time period than the previous drying stage (420 min < 360 min < 240 min < 180 min < 120 min). The moisture reduction during the last time interval of hot-air drying (360 to 420 min) at 60 ° was 1.16%, which exhibited minimum evaporation of water from mushroom tissues. Furthermore, the small difference of mean intensities between 360 min and 420 min of drying assumed that bound water transfers into certain amount of nonfreezable water that exists in the intracellular space of the mushroom (see Figure 3). Wu et al. (2012) used spectroscopy techniques from the VIS region to the NIR for the inspection of water contents of food. As a consequence of O–H third stretching overtone absorption of waveband between 940 and 970 nm which was certainly allocated moisture contents (Wu, He, & Feng, 2008). Spectral amplitude is directly proportional to the moisture contents, such as fresh samples produced highest reflected spectra because of the presence of high water contents. The increment of drying time decreases moisture contents as well as decreases mean intensity. Spectral reflectance tendency of mushroom strengthens the concept that higher the moisture contents, greater the mean intensity outcomes in the region of 660 to 870 nm during the dehydration of carrot slices (C. Liu et al., 2016) and nondestructive inspection of water contents in beef (J. Liu et al., 2016).

3.2 Influence of hot-air drying on water mobility

Heat process causes considerable minor or major changes in the microstructure of food material. The alterations in textural properties are the consequences of heat treatment as well as water transportation throughout the structure. The evaporation of free water resulted in surface discoloration, whereas the removal of bound water caused structural shrinkage. LF-NMR was used for

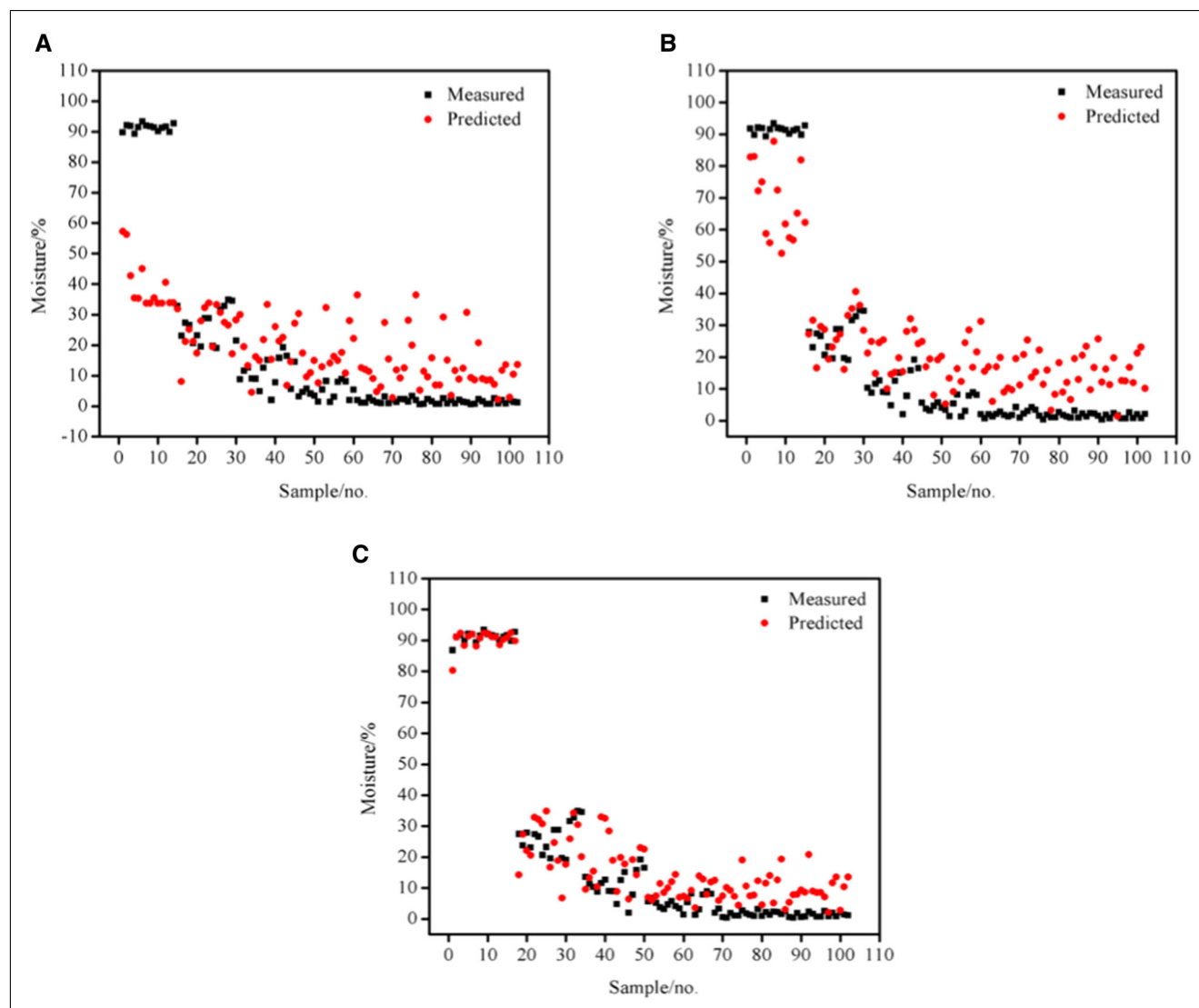


Figure 5—Presentation of quantitative models: (A) LS-SVM, (B) PLS, and (C) BPNN.

the investigation of water status of hot-air dried shiitake mushrooms for six different time periods through T_2 relaxation time. Multiple curve fitting was performed, which separates the peaks. The mobility of water comprised of transverse relaxation times T_{21} (<10 ms) T_{22} (10 to 100 ms), and T_{23} (100 to 1,000 ms) indicates specific water form of bound, immobilized, and free water, respectively presented in Figure 4A (Lu et al., 2017; Xu, Jin, Zhang, & Chen, 2017). A total of $89.66\% \pm 2.94\%$ water evaporated from shiitake cubes. Initially higher water evaporation rate was observed such as $60.37\% \pm 2.39\%$ water evaporated in 120 min, whereas only $29.29\% \pm 0.68\%$ was removed in next 300 min because maximum water existed in free and immobilized forms (see Figure 3). Change of relaxation time with respect of time showed rapid decline of free and immobilized water. Peak's height of each relaxation curve Figure 4(A) explains the trend of water forms linked with heat operation and describes the movement from higher to lower relaxation time inside of food structure. The signal intensity of T_{23} showed whole free water removed within 240 min of drying. The signal intensity of T_{21} acquired high values indication of increasing proportion of bound. However, peak height of T_{21} after 360 min of heat operation exhibited the pres-

ence of bound water at this stage and no major changes of water due to strong microstructural bonds. Subsequently proceeding of heat process for 420 min removed $89.66\% \pm 2.94\%$ moisture contents, and small reduction was observed in T_{21} amplitude. This reduction illustrates the significant effect of hot-air drying on free and immobilized water forms while insignificant for the removal of bound water contents of mushroom cubes. Previous research of NMR on bound water showed similar nonsignificant results of hot-air drying in broccoli (Xu et al., 2017). Figure 4(B) illustrated the comparison of percentage of each water fraction. Raw mushroom cubes contained $46.46\% \pm 0.01\%$, $50.53\% \pm 0.02\%$, and $3.01\% \pm 0.02\%$ of free, immobilized, and bound water, respectively, out of $91.03\% \pm 1.93\%$ of total moisture. It means maximum water existed in intracellular spaces. Halder, Datta, and Spanswick (2011) proposed similar results that maximum amount of water presents in intracellular spaces of apple tissues. The 240 min of drying process resulted in maximum proportion of bound water ($93.44\% \pm 0.40\%$) out of $8.68\% \pm 4.74\%$ of total moisture. Further drying till 420 min showed decrease trend of bound water and rise in immobilized water (Figure 4B). The phenomenon of conversion and migration of bound into immobilized water was due

to the ruptured cell membrane (Khan, Wellard, Nagy, Joardder, & Karim, 2017).

3.3 Quantitative analysis of different models to investigate water status

The descriptive statistics of changes in moisture contents are of two sets (calibration and prediction; presented in Table 1). Different drying period produced variability of moisture contents from 93.86% to 0.28% of sets. As confirmed by water status, most of the free and immobilized water was removed after 240 min of dehydration during hot-air drying, while only bound water remained (see Figure 4). Drying approach produced large variation in spectral data within fixed time periods and yielded $1.38\% \pm 0.73\%$ final moisture contents. This spectral information was divided into calibration and prediction sets and evaluated by quantitative models developed through mathematical algorithms PLS, BPNN, and LS-SVM. Based on these three models' statics, capability of MSI predicts the water status presented in Table 2. Various parameters were involved in the model development step such as LS-SVM with RBF combination of (γ , σ^2) considered at the value of (256, 0.0039) to predict moisture. Same as earlier, LVs were counted on the basis of 10 with minimum PRESS value for PLS model development. According to the values of coefficient of determination (R^2) for calibration and prediction sets, BPNN was found to be the most precise model to determine the water status on every stage of the process with highest $R_c^2 = 0.9829$, $R_p^2 = 0.9639$, and lowest RMSEC and RMSEP values of 0.2289% and 0.2291%. However, the PLS model showed results better than LS-SVM. The quantitative models for prediction of water status are presented in Figure 5. The investigation of moisture contents of carrot slices through hot-air drying showed similar results while model establishment (C. Liu et al., 2016). However, BPNN is relatively an excellent approach to observe water status and PLS was comparatively better than LS-SVM with lower RMSEC and RMSEP of 8.8511 and 15.3749, respectively. In consequence of the above discussion, BPNN was conceded as the best established model to quantify water status of shiitake mushroom during drying process.

4. CONCLUSION

The findings of this study lead us to effectively monitor water forms during the dehydration process by the utilization of non-destructive methodology and exploit the intensive application of MSI in the food sector. We used LF-NMR method as a reference study in consequences of MSI evaluation. The LF-NMR study exhibited that mushrooms contained maximum amount of immobilized water followed by free water. These water types could be easily removed in a short time of 240 min as well as more efforts were required to remove bound water if required. The utilization of videometer vision system for images acquisition (VIS-NIR region: 405 to 970 nm) is considered a promising inspection of quality attributes of water. MSI combined with chemometrics is a promising technique to discriminate water status in a fast and nondestructive way. BPNN proved to be the best predictive model with highest accuracy of $R_p^2 = 0.9639$, RMSEP = 0.2291 as compared with LS-SVM and PLS models. Moreover, the results highlighted the aspect of MSI, providing both spectral and spatial information together is beneficial for online assessment of water status, which could overcome the difficulty caused by the migration and conversion of different water phases during the drying process. The nondestructive approach will also be considered as a best strategy in future to investigate mushrooms under different conditions in food processing industry.

ACKNOWLEDGMENTS

We acknowledge the National Key Research and Development Program of China (2016YFD0401104) and the Key Science & Technology Specific Projects of Anhui Province (15cz03117, 16030701078) for supporting this work, and also acknowledge the Funds for Huangshan Professorship of Hefei University of Technology (407-037019).

CONFLICTS OF INTEREST

There are no conflicts of interest to declare.

AUTHOR CONTRIBUTIONS

Shoaib Younas: He has done the main part of the experiments and declared as main author.

Changhong Liu and Wei Liu: Their contribution is to develop the prediction models.

Hao Qu: The contribution from this author is the experiment design.

Liyang Wei and Ling Yan: They put effort to do experiments and data analysis.

Yu Mao: Supervised the experimental results and technical paper writing.

Lei Zheng: Providing the research idea and supervision of the whole project.

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